

Review article

Effectiveness and complication rates of high intensity focused ultrasound treatment for abdominal wall endometriosis: A systematic review

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ABSTRACT

In this review, a systematic literature search on the effectiveness and complication rates of ultrasound-guided and magnetic resonance-guided high-intensity focused ultrasound (USg-/MRgHIFU) for abdominal wall endometriosis (AWE) was conducted in six databases in May/June 2023. Original articles of (non)randomized trials, cohort studies, case-control studies and case series published in peer-reviewed journals were included. Of the included studies the level of evidence (LoE) and methodological quality using the ROBINS-I and IHE-QAT was assessed. Primary outcomes were non-perfused volume ratio (NPV%), lesion size, pain scores, side effects and complication rates according to Society of Interventional Radiology (SIR) guidelines. Secondary outcomes were recurrence and re-intervention rates.

Seven cohort studies (one of good methodological quality) (LoE 3) on USgHIFU were included ($n = 212$, AWE lesions = 240–245). Six months after USgHIFU treatment, pain scores were reduced with 3.3–5.2 points (baseline: 5.1–6.8, $n = 135$). Self-limiting side effects were pain (85.7 % (114/133)) and swelling (34.6 % (46/133)) in the treatment area. Complications occurred in 17.7 % (32/181), all of which were minor. Recurrence occurred in 12.8 % (11/86). Three of these seven cohort studies compared USgHIFU ($n = 61$) with surgical excision ($n = 74$). Pooled results showed no significant differences in pain scores, complications (resp. 26.3 % (10/38) vs. 32.6 % (15/46) ($p = 0.53$)) and recurrences (resp. 4.9 % (3/61) vs. 5.4 % (4/74) ($p = 0.90$)).

This systematic review suggests that HIFU is an effective and safe treatment option for AWE. USgHIFU treatment led to reduced pain scores and lesion size, was free of major complications and had a pooled recurrence rate of 12.8 %. Compared to surgical excision pooled results showed no significant differences in pain scores, complications and recurrences after USgHIFU. However, many of the included studies had limitations in their methodological quality and therefore the results should be interpreted with caution. Well-structured high-quality randomized controlled trials comparing HIFU to standard care should be conducted to provide more conclusive evidence.

Introduction

Abdominal wall endometriosis (AWE) is the presence of endometrium-like epithelium or stroma within the abdominal wall. AWE develops in 80 % of cases near the incision site after gynecological surgery, mostly cesarean sections (CSs). It develops months to years after a CS in 0.03–0.45 % [1–4]. In the Netherlands, with a CS rate of 18

%, this corresponds to 9–138 cases annually [5].

The exact pathogenesis remains unclear. The prevailing theory implicates the introduction of viable endometrial tissue in the incision during surgery, which proliferates under hormonal influences, causing endometriosis [3,6]. Clinical presentation varies, with a classic triad of a palpable mass, cyclic pain and history of CS.

Pharmaceutical treatment with hormonal therapy and analgesics is often insufficient [7]. The only definitive treatment option is wide

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Abbreviations

Abdominal wall endometriosis AWE
 Cesarean section CS
 Contrast-enhanced ultrasound CEUS
 High-intensity focused ultrasound HIFU
 Magnetic resonance-guided high-intensity focused ultrasound MRgHIFU
 Non-perfused volume NPV
 Non-perfused volume ratio NPV%
 T1-weighted contrast-enhanced T1w CE
 Ultrasound-guided high-intensity focused ultrasound USgHIFU
 Ultrasound US

surgical excision. This can be extensive, requiring mesh repairs (6.0–30 %), and carries a risk of complications as hematomas or seromas (10.8 %), postoperative fever (3.0 %) and postoperative hernias (18.2 %). Recurrences occur in 12.5–28.6 % after incomplete surgical excision, especially when lesions infiltrate the rectus abdominis muscle [3,8].

High Intensity Focused Ultrasound (HIFU) may be a suitable alternative to surgical excision. In HIFU focused high-intensity ultrasound (US) beams create a focal point wherein temperature rises above 55 °C. This thermally ablates the targeted tissue resulting in shrinkage. Near or far field heating around the targeted area is inevitable, which possibly, but rarely, causes complications. HIFU treatment is performed under real-time guidance using ultrasound guided or magnetic resonance guided HIFU (USg-/MRgHIFU) [9,10] Immediately after HIFU, technical success is assessed by determining the non-perfused volume ratio (NPV %): the ratio of post-treatment non-perfused volume (NPV) and pre-treatment lesion volume.

Due to its non-invasive nature and low complication rates, as demonstrated in the treatment of uterine fibroids and adenomyosis, HIFU might be beneficial in the treatment of AWE, especially with the expected increase in AWE incidence due to increasing CS rates worldwide and increasing BMI among women undergoing CS [1,3,11,12].

The objective of this study was to systematically review current literature on the effectiveness and complication rates of HIFU for AWE. Secondary outcomes were recurrence and re-intervention rates.

Methods

This review was conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) 2020 statement. (Supplementary material 1) [13] The review was registered in the International Prospective Register of Systematic Reviews (PROSPERO) with registration number CRD42024499583.

Data sources and study selection

Two researchers (LK, LR) independently searched six databases in May/June 2023: Cochrane Library, PubMed/MEDLINE, Embase, Trip, ClinicalTrials.gov and the International Clinical Trials Registry Platform. See Supplementary material 2 for the search strategy. We included (non)randomized trials, cohort studies, case-control studies and case series published in peer-reviewed journals, that evaluated USg-/MRgHIFU treatment in patients with isolated symptomatic AWE.

Articles were excluded if none of the primary outcomes were

described or when the full-text article was not available in English. References of included articles were searched for additional publications. Disagreements regarding inclusion were resolved by consensus. If consensus could not be reached, a third researcher was consulted (IN).

Data extraction

Two researchers (LK, LR) independently extracted the following data from included articles: study design, follow-up time, age, BMI, history of gynecological surgery, NPV%, pain scores and lesion size before and after treatment including imaging modality, side effects, complications, and recurrence and re-intervention rates.

Effectiveness was measured in terms of NPV%, pain reduction (based on pain scores from zero to ten on pain rating scales) and reduction in AWE lesion size.

A distinction was made between side effects and complications. Side effects were defined as expected symptoms, e.g. mild pain or discomfort during HIFU treatment, without the need for additional therapy or discontinuation of treatment. Complications were defined as unexpected symptoms and were categorized according to the Society of Interventional Radiology (SIR) clinical practice guidelines. SIR A and B complications were considered minor, SIR C, D, E and F major [14].

Quality assessment

Level of evidence (LoE) was determined independently by two researchers (LK, LR) using the Oxford Centre for Evidence-based Medicine (OCEBM) guidelines, which are used to find the most reliable evidence to answer a clinical problem. Level one indicates the highest LoE, level five the lowest [15].

Methodological quality was assessed independently by two researchers (LK, LR) using the Risk Of Bias In Non-randomized Studies - of Interventions (ROBINS-I) tool for nonrandomized comparative studies, and the Institute of Health Economic Quality Appraisal Tool (IHE-QAT) for non-comparative studies [16–18]. The question in the IHE-QAT whether patients entered the study at a similar stage of the disease, was excluded, since this is not applicable for AWE. Disagreements were resolved by discussion. If consensus could not be reached, a third researcher was consulted (IN). The ROBINS-I classified methodological quality as low, moderate, serious or critical risk of bias. The IHE-QAT indicated good methodological quality with a score $\geq 13/19$.

Missing data

When the number of AWE lesions per patient was not reported, it was assumed that one lesion per patient was treated. When median (m), range (a-b) and sample size (n) were reported the mean ($\bar{X}$) and standard deviation (SD) were estimated according to these formulas [19]:

$$\bar{X} \approx ((a + 2m + b)/4) + ((a - 2m + b)/4n)$$

$$SD \approx \begin{cases} 1/\sqrt{12} * [(b - a)^2 + (((a - cm + b)^2)/4)]^{1/2} & n \leq 15 \\ (b - a)/4 & 15 < n \leq 70 \\ (b - a)/6 & n > 70 \end{cases}$$

NPV% was calculated using this formula:

$$NPV\% = (Non - perfused volume immediately after HIFU (cm^3) / Lesion size before HIFU (cm^3)) * 100\%$$

Sometimes the calculated NPV% differed from the reported NPV%. In this case, we chose to report our calculated NPV%.

AWE lesion size on follow-up was reported as volume reduction (%) using this formula: $\text{Volume reduction} = \frac{\text{Lesion size at follow-up (cm}^3 \text{ or cm)}}{\text{Lesion size at baseline (cm}^3 \text{ or cm)}} \times 100\%$

Change scores of pain scores were calculated as follows:

$$\text{Change score} = \text{pain score}_{\text{baseline}} - \text{pain score}_{\text{final}}$$

The SDs of change scores and volume reduction were calculated with a correlation coefficient of 0.80 using this formula:

$$SD_{\text{change}} = \sqrt{SD_{\text{baseline}}^2 + SD_{\text{final}}^2 - (2 * \text{Correlation coefficient} * SD_{\text{baseline}} * SD_{\text{final}})}$$

Statistics

A reliable meta-analysis or meta-regression could not be performed due to the small sample size and heterogeneity of included studies.

Ethical approval

Ethical Review Board approval was not required as all data was extracted from published articles.

Results

Literature search

The literature search identified 304 potentially relevant articles.

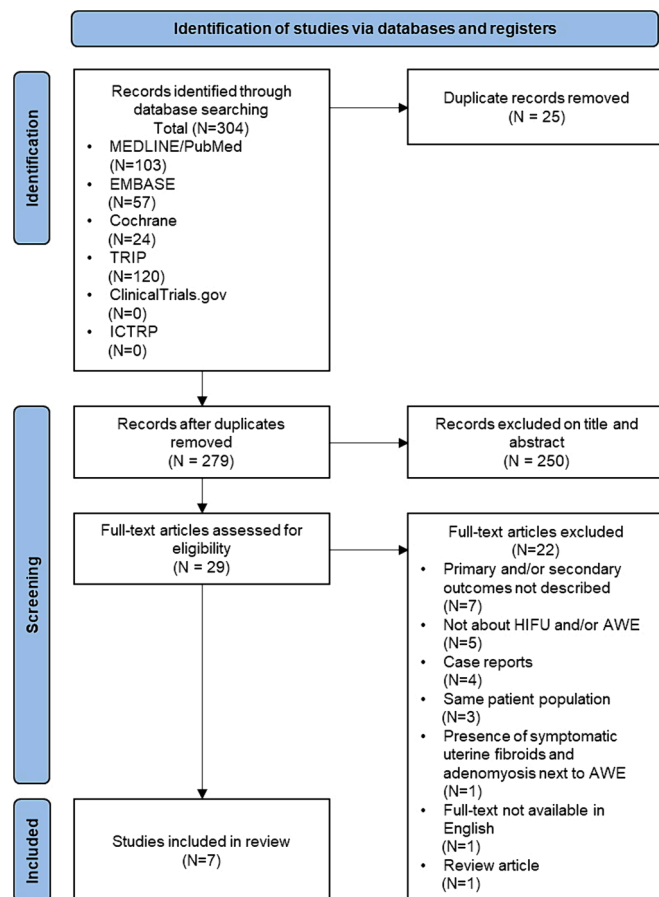


Fig. 1. PRISMA flow chart. Flow chart of results of systematic literature search in several databases and in- and exclusion of articles according to PRISMA guidelines.

After removing duplicates, 279 articles were screened for eligibility on title and abstract, resulting in the screening of 29 full-text articles. Four articles appeared to report at least partially on the same patient population, of which the study with the largest numbers of patients was included [20–23]. Seven articles met the eligibility criteria for final inclusion [23–29] (Fig. 1).

Study characteristics

All articles were cohort studies, six retrospective. In one article, it was unclear whether the study was retrospective or prospective [29]. Three studies of the seven included studies compared USgHIFU treatment with surgical excision, which is evaluated in the *Comparison with surgery* section [24,25,27].

The last follow-up ranged from 1 to 46 months. Four studies provided results from several follow-up points, i.e. at one, three, six or twelve months after USgHIFU [23,24,26,27] (Table 1).

Quality of evidence

The LoE of all studies was three. According to the ROBINS-I, all three comparative studies had a serious risk of bias, mostly due to confounding. Among the non-comparative studies, only one study was of good methodological quality (Table 1) (Supplementary material 3).

Characteristics of patients and AWE lesions

Of all patients (n = 210), 195 (92.9 %) had undergone prior gynecological surgery. A single AWE lesion was present in 185 patients, two lesions in 13 patients and three lesions in 7 patients. One study (n = 25) reported that five patients had two or three lesions, but did not specify how many lesions were treated [25]. Therefore, we applied a range of 238–243 AWE lesions. One study (n = 23) did not describe the involvement of structures [27]. Of the remaining 215–220 lesions (n = 187) subcutaneous fat tissue was involved in 123, rectus muscle in 80, subcutaneous fat tissue and rectus muscle in 69, fascia of the rectus muscle in 4 and peritoneum in 1 (Table 2).

HIFU treatments and NPV%

All studies performed USgHIFU on a Chongqing system (Haifu Medical Technology, Chongqing, China) (n = 210, (range n per study: 13–67)) (Table 3).

NPV% could be calculated in three studies. Wang et al. [23] (n = 67) and Luo et al. [26] (n = 31) used contrast-enhanced US (CEUS) to determine AWE lesion volume before and NPV after USgHIFU treatment, Hu et al. [29] used T1-weighted contrast-enhanced MRI. This resulted in NPV% of 177 ± 233 %, 58 ± 17 % and 95 ± 82 % respectively. The other studies did not report NPV or NPV% and these could not be calculated due to lack of data.

Pain scores

Pain scores were reported in five studies. The follow-up period was three to twelve months, except for one study. In this study (n = 25), the baseline numeric rating scale (NRS) score was 6.9, which decreased to 0.3 at the last follow-up (median 32 months (range: 19–46)) [25].

Three studies (n = 103) reported visual analogue scale (VAS) scores at one, six and twelve months after USgHIFU, all showing stabilization or progressive reduction in pain scores (range means: 5.1–5.8, 1.2–2.0, 0.6–1.8 and 0.5–1.8 respectively) [23,24,27]. One study (n = 67) reported a statistically significant difference in the VAS scores between baseline (5.5 ± 0.7) and six months (1.0 ± 1.4) and baseline and twelve months (0.5 ± 0.7) after USgHIFU (both $p < 0.01$) [23]. Another study (n = 32) reported a statistically significant ($p < 0.05$) reduction in pain scores from 6.8 ± 2.6 at baseline to 1.8 ± 0.3 six months after USgHIFU

Table 1
Characteristics of included studies.

Study	Study design	Country	Inclusion	Number of patients (n)	Number of AWE lesions (n)	Control group (n)	USg/MRg	Outcomes	FU (months)	LoE	Quality
Wang et al, 2021	Retrospective cohort	China	AWE	67	82	No	USg	NPV/NPV% VAS Lesion volume AE/Complications	12	III	13/19*
Shi et al, 2020	Retrospective cohort	China	AWE	13	13	Surgery (17)	USg	VAS Lesion volume AE/Complications Recurrence rate	12–24	III	Serious risk of bias**
Zhao et al, 2018	Retrospective cohort	China	AWE	25	30–35	Surgery (29)	USg	Technical success NRS AE/Complications Recurrence rate	19–46	III	Serious risk of bias**
Luo et al, 2017	Retrospective cohort	China	AWE	32	39	No	USg	NPV Cyclic pain rating scale Lesion volume AE/Complications	12	III	12/19*
Zhu et al, 2017	Retrospective cohort	China	AWE	23	23	Surgery (28)	USg	VAS Lesion diameter Days of hospital stay AE/Complications Recurrence rate	1–12	III	Serious risk of bias**
Wang et al, 2011	Retrospective cohort	China	AWE	21	21	No	USg	Presence of cyclic pain Presence of lesion at FU AE/Complications	3–31	III	11/19*
Hu et al, 2022	Retrospective/prospective cohort	China	AWE	29	30	No	USg	MRI and US characteristics NPV/NPV% Recurrence rate	LOD	III	12/19*

AE = adverse events, AWE = abdominal wall endometriosis, FU = follow-up, LoE = level of evidence, MRg = magnetic resonance-guided, NPV = non-perfused volume, NPV% = non-perfused volume ratio, NRS = numeric rating scale, USg = ultrasound-guided, VAS = visual analogue scale.

* Assessment with IHE-QAT. A score $\geq 13/19$ points indicates good methodological quality.

** Assessment with ROBINS-I. Risk of bias is classified as low, moderate, serious or critical.

(pain scale not specified). [26] (Table 4) (Fig. 2).

AWE lesion size

In three studies (n = 112, AWE lesions = 134) lesion size was displayed as volume (cm³) measured using CEUS before and after USgHIFU [23,24,26]. Mean baseline volumes were 10.3 ± 12.8 cm³ (n = 67), 2.3 ± 2.1 cm³ (n = 13), and 2.8 ± 0.1 cm³ (n = 32). At follow-up, lesion volume was measured using color Doppler US, US or MRI, or the imaging technique was not specified. These three studies reported a mean volume reduction six months after USgHIFU treatment of 81.9 ± 15.7 % (n = 61, imaging technique not specified), 60.8 ± 22.2 % (n = 13, MRI or color Doppler US) and 52.5 ± 11.1 % (n = 32, US or MRI), respectively. Two of these three studies showed continuous volume reduction, i.e. from 81.9 ± 15.7 % to 96.2 ± 5.4 % (n = 58, imaging technique not specified) and from 60.8 ± 22.2 % to 70.3 ± 20.9 % (n = 13, US or MRI) at twelve months follow-up [23,24]. The other study had no follow-up at twelve months [26].

One study (n = 23, AWE lesions = 23), displayed lesion size as maximum diameter (cm) before and after USgHIFU [27]. Baseline maximum diameter measured using CEUS was 2.7 ± 1.3 cm. At follow-up, maximum diameter was measured using color Doppler US and showed a reduction of 44.1 ± 14.8 % and 49.0 ± 17.6 % at one and twelve months after treatment respectively. The three remaining studies did not report on lesion size at follow-up. (Table 5) (Fig. 3).

Side effects and complication rates

Side effects could be extracted from four studies (n = 133)

[23,24,26,28]. The other studies did not report side effects. Pain in the treatment area during or after USgHIFU was reported in 85.7 % of patients (114/133). One study (n = 67) reported VAS scores during and immediately after treatment of 5.5 ± 0.7 and 1.0 ± 1.4 respectively [23]. The other studies did not report pain scores immediately after treatment but mentioned that pain was tolerable and did not require analgesics. Mild swelling in the treatment area immediately after USgHIFU treatment occurred in 34.6 % of patients (46/133) and resolved spontaneously after 1–3 days.

Six studies reported complications (n = 181) [23–28]. Only minor complications occurred. SIR A complications occurred in 16.0 % (29/181), and included skin thermalgia (6.6 % (12/181)), hematuria (5.5 % (10/181)) and orange peel skin appearance (3.9 % (7/181)). SIR B complications occurred in 1.7 % of patients (3/181), which were all skin burns. (Table 6).

Recurrence and re-intervention rates

Four studies reported 11 recurrences in 86 patients (12.8 %) [24,25,27,29]. The other studies did not report recurrences. Hu et al. [29] (n = 25) reported most recurrences (n = 8), but follow-up time was not reported. They showed significantly more recurrences in the NPV < 50 % group (range: 0–46.8 %) (80 % (4/5)) compared to the NPV > 50 % group (range: 50.1–318.2 %) (20 % (4/20)) (p < 0.05). Follow-up in the other three studies ranged from 12 to 46 months. None of the articles reported re-interventions.

Table 2
Characteristics of patients and AWE lesions.

Study	Number of patients (n)	Age (years) (mean ± SD)	BMI (kg/m2) (mean ± SD)	Surgical history(n(%))	Number of AWE lesions(n(%))	Involved structures of AWE lesion(n(%))		
Wang et al, 2021	67	34.0 ± 2.8	26.9 ± 6.2	CS – 1 – 2 Unclear	52 (77.6) 45 (67.2) 7 (10.4) 15 (22.4)	82 – 1: 54 (65.9) – 2: 7 (8.5) – 3: 6 (7.3)	- SFT and RM - SFT - Fascia	63 (76.8) 15 (18.3) 4 (4.9)
Shi et al, 2020	13	31.2 ± 4.2	21.8 ± 3.5	CS	13 (100)	13	- SFT - Anterior sheath RM - RM	10 (80)1 (8)2 (15)
Zhao et al, 2018	25	31.8 ± 4.3	21.1 ± 2.7	CS – 1 – 2	25 (100) 22 (88)3 (12)	30–35 – 1: 20 (80) – 2–3: 5 (50)	- SFT - Anterior sheath RM - Muscular coat	21 (84)19 (76)9 (36)
Luo et al, 2017	32	39.4 ± 6.9	LOD	CS Hysterectomy	31 (97)1 (3)	39 – 1: 26 (81) – 2: 5 (16) – 3: 1 (3)	- SFT - RM - Peritoneum	24 (62)14 (36)1 (2.6)
Zhu et al, 2017	23	30.1 ± 4.9	LOD	CS	23 (100)	23	LOD	
Wang et al, 2011	21	33.5 ± 5.1	LOD	CS Hysterectomy	20 (95)1 (5)	21 – 1: 21 (100)	- SFT - RM - SFT and RM	3 (14)12 (57)6 (29)
Hu et al, 2022	29	32.5 ± 4.5	LOD	CS – 1 - ≥2	29 (100) 28 (96.6)1 (3.4)	30 – 1: 28 (96.6) – 2: 1 (3.4)	- SFT - Anterior sheath RM - Posterior sheath RM	5 (14.4) 15 (50)10 (36.7)

AWE = abdominal wall endometriosis, BMI = body mass index, CS = cesarean section, LOD = lack of data, NPV% = non-perfused volume ratio, RM = rectus abdominis muscle, SD = standard deviation, SFT = subcutaneous fat tissue.

Table 3
Technical characteristics of USgHIFU treatment.

Study	Number of patients (n)	USgHIFU system (model)	Sonication time (s) (mean ± SD)	Treatment time (min) (mean ± SD)	Power (W) (range)	Thermal dose (J) (mean ± SD)	NPV% (%) (mean ± SD)
Wang et al, 2021	67	Chongqing (JC)	371 ± 29.70	64 ± 37	100–300	49,900 ± 12,587	177 ± 233
Shi et al, 2020	13	Chongqing (JC)	315 ± 150	13 ± 11	100-LOD	LOD	LOD
Zhao et al, 2018	25	Chongqing (JC)	287 ± 211	60 ± 64	100–200	LOD	LOD
Luo et al, 2017	32	Chongqing (JC200)	252 ± 75	98 ± 15	5–150	LOD	58 ± 17
Zhu et al, 2017	23	Chongqing (JC200)	322 ± 190	71 ± 12	100–300	LOD	LOD
Wang et al, 2011	21	Chongqing (JC)	LOD	20 ± 11	200–400	LOD	LOD
Hu et al, 2022	29	Chongqing (JC)	LOD	LOD	LOD	LOD	95 ± 82

LOD = lack of data, NPV% = non-perfused volume ratio, SD = standard deviation, USgHIFU = ultrasound guided high-intensity focused ultrasound.

Comparison with surgery

Three studies compared USgHIFU (n = 61, AWE lesions = 66–71) with surgical excision (n = 74) [24,25,27] One study reported significantly shorter mean treatment times in the USgHIFU group (13 ± 11 min, n = 13) compared to the surgery group (45 ± 5 min, n = 17) (p < 0.01) [24]. Another study reported no statistical difference in mean treatment time, 60 ± 64 (n = 25) and 56 ± 46 min (n = 29), respectively (p = 0.78) [25] In all studies, mean hospital stay was significantly shorter after USgHIFU (range means: 1.0–5.0 days) than after surgery (range means: 5.2–7.5 days) (all p < 0.05).

One study reported NRS scores immediately after treatment, which were significantly lower in the USgHIFU group (3.4 ± 1.2, n = 25) than in the surgery group (5.5 ± 1.5, n = 28) (p < 0.001) [25]. In the other studies (USgHIFU n = 36, surgery n = 45), no significant difference was

reported in VAS scores before and one, three, six and twelve months after treatment between the groups [24,27].

One study (USgHIFU n = 13, surgery n = 17) reported side effects within 24 h after treatment, showing no difference in presence of pain at the USgHIFU treatment area (61.5 %) and the surgical incision site (58.8 %) (p = 0.60) [24].

Two of three studies reported complications, which occurred in 26.3 % (10/38) of USgHIFU patients and 32.6 % (15/46) of surgery patients (p = 0.53) [24,25]. Only minor complications occurred in the USgHIFU group (n = 38). Due to lack of information on the treatment of complications in surgery patients, the distinction per SIR classification could not be made. One study mentioned that four complications in the surgical group (n = 17) were major [24].

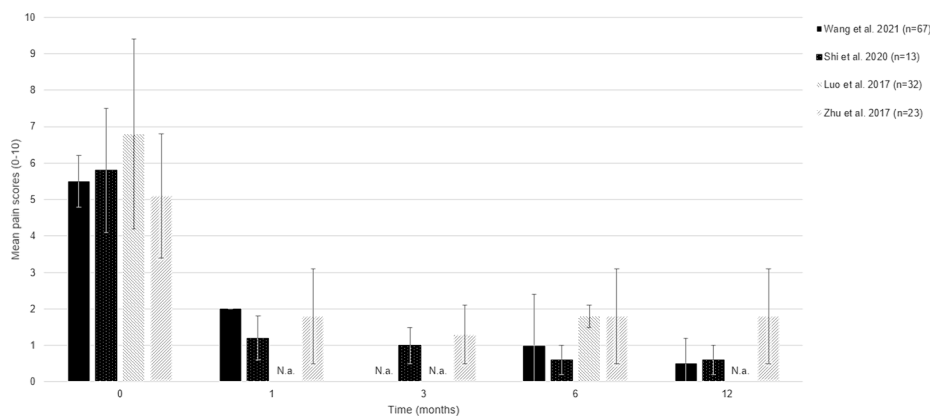
Overall, recurrence rates in USgHIFU and surgery groups were 4.9 % (3/61) and 5.4 % (4/74) respectively (p = 0.90).

Table 4

Pain scores at 1, 3, 6 and 12 months after USgHIFU.

Study	Number of patients (n)	Pain rating scale	Baseline	1 month FU (mean ± SD)		3 months FU		6 months FU		12 months FU	
			Pain score (mean ± SD) (N)	Pain score (mean ± SD) (N)	Change score (mean ± SD)	Pain score (mean ± SD) (N)	Change score (mean ± SD)	Pain score (mean ± SD) (N)	Change score (mean ± SD)	Pain score (mean ± SD) (N)	Change score (mean ± SD)
Wang et al, 2021	67	VAS	5.5 ± 0.7 (67)	2.0 ± 0.0 (67)	3.5 ± 0.7	LOD	LOD	1.0 ± 1.4 (61)*	4.5 ± 0.9	0.5 ± 0.7 (58)*	5.0 ± 0.6
Shi et al, 2020	13	VAS	5.8 ± 1.7 (13)	1.2 ± 0.6 (13)	4.6 ± 1.4	1.0 ± 0.5 (13)	4.8 ± 1.4	0.6 ± 0.4 (13)	5.2 ± 1.5	0.6 ± 0.4 (13)	5.2 ± 1.5
Zhao et al, 2018	25	NRS	6.9 (25)	LOD	LOD	LOD	LOD	LOD	LOD	LOD	LOD
Luo et al, 2017	32	LOD	6.8 ± 2.6 (32)	LOD	LOD	LOD	LOD	1.8 ± 0.3 (32)**	5.0 ± 2.5	n.a.	n.a.
Zhu et al, 2017	23	VAS	5.1 ± 1.7 (23)	1.8 ± 1.3 (LOD)	3.3 ± 0.8	1.3 ± 0.8 (LOD)	3.8 ± 1.2	1.8 ± 1.3 (LOD)	3.3 ± 0.8	1.8 ± 1.3 (LOD)	3.3 ± 0.8

FU = follow-up, LOD = lack of data, n.a. = not applicable, NRS = numeric rating scale, SD = standard deviation, VAS = visual analogue scale.

* significant ($p < 0.01$), ** significant ($p < 0.05$).**Fig. 2.** Pain scores after USgHIFU for AWE. Mean pain scores with standard deviations at baseline, 1, 3, 6 and 12 months after ultrasound guided high intensity focused ultrasound (USgHIFU) of abdominal wall endometriosis (AWE) lesion. n = number of participants at baseline n.a. = not applicable: data is missing in Wang et al. 2021 3 months follow-up and in Luo et al. 2017 at 1, 3 and 12 months follow-up.**Table 5**

Lesion size before USgHIFU and reduction of lesion size at 1, 3, 6 and 12 months after USgHIFU, including imaging modality.

Study	Number of patients (n)	Baseline	1 month FU	3 months FU	6 months FU	12 months FU
		Max. diameter (cm) or volume(cm ³) (mean ± SD) (n) (imaging tool)	Size reduction (%) (mean ± SD) (n) (imaging tool)	Size reduction (%) (mean ± SD) (n) (imaging tool)	Size reduction (%) (mean ± SD) (n) (imaging tool)	Size reduction (%) (mean ± SD) (n) (imaging tool)
Wang et al, 2021	67	10.3 ± 12.8 cm ³ (67) (CEUS)	16.7 ± 6.6 (67) (LOD)	LOD	81.9 ± 15.7* (61) (LOD)	96.2 ± 5.4* (58) (LOD)
Shi et al, 2020	13	2.3 ± 2.1 cm ³ (13) (CEUS)	47.8 ± 43.9 (13) (MRI or CDUS)	56.0 ± 37.0 (13) (MRI or CDUS)	60.8 ± 22.2 (13) (MRI or CDUS)	70.3 ± 20.9 (13) (MRI or CDUS)
Zhao et al, 2018	25	3.0 ± 1.2 cm ³ (25) (BMUS)	n.a.	n.a.	n.a.	n.a.
Luo et al, 2017	32	2.8 ± 0.1 cm ³ (32) (CEUS)	LOD	LOD	52.5 ± 11.1 (32)** (US or MRI)	n.a.
Zhu et al, 2017	23	2.7 ± 1.3 cm (23) (CEUS)	44.1 ± 14.8 (LOD) (CDUS)	41.6 ± 15.7 (LOD) (CDUS)	45.3 ± 15.7 (LOD) (CDUS)	49.0 ± 17.6 (LOD) (CDUS)
Wang et al, 2011	21	2.8 ± 1.1 cm (21) (CDUS)	LOD	LOD	LOD	LOD
Hu et al, 2022	29	3.3 cm ³ ± 1.5 (29) (MRI)	n.a.	n.a.	n.a.	n.a.

AWE = abdominal wall endometriosis, BMUS = B-mode ultrasound, CDUS = color Doppler ultrasound, CEUS = contrast-enhanced ultrasound, FU = follow-up, LOD = lack of data, MRI = magnetic resonance imaging, n.a. = not applicable, SD = standard deviation, US = ultrasound, USgHIFU = ultrasound-guided high-intensity focused ultrasound.

* significant ($p < 0.01$), ** significant ($p < 0.05$).

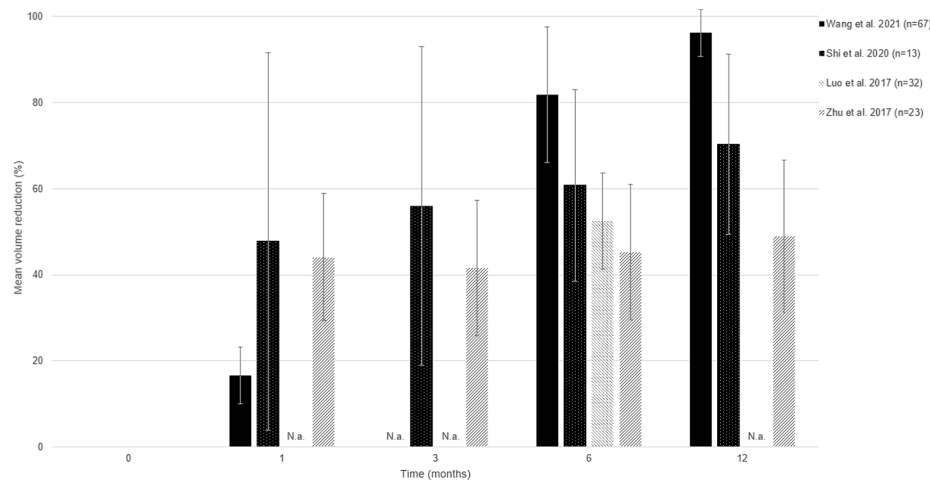


Fig. 3. Reduction in AWE volume or lesion size after USgHIFU. Mean reduction in volume or lesion size (%) with standard deviations of abdominal wall endometriosis (AWE) lesion after ultrasound guided high intensity focused ultrasound (USgHIFU). n = number of participants at baseline. n.a. = not applicable, data is missing in Wang et al. 2021 at 3 months follow-up and in Luo et al. 2017 at 1, 3 and 12 months follow-up.

Table 6
Side effects and complications after USgHIFU according to SIR classification.

Study	n	Side effects (n(%))		SIR A (n(%))		SIR B (n(%))	
Wang et al, 2021	67	Pain in treatment area	67 (100)	Skin thermalgia	12 (17.9)7	Skin burn	2 (3.0)
				Hematuria	(10.4)2		
				Orange peel skin appearance	(3.0)		
Shi et al, 2020	13	Pain in treatment area	8 (61.5)	Hematuria	1 (7.7)	Total	0 (0)
Zhao et al, 2018	25	LOD	LOD	Orange peel skin appearance	5 (20.0)	Skin burn	1 (4.0)
				Hematuria	2		
					(8.0)		
Luo et al, 2017	32	Pain in treatment area	32 (100)25	Total	0 (0)	Total	0 (0)
		Swelling in treatment area	(78.1)				
Zhu et al, 2017	23	LOD	LOD	Total	0 (0)	Total	0 (0)
Wang et al, 2011	21	Pain in treatment area Swelling in treatment area	7 (33.3)21	Total	0 (0)	Total	0 (0)
			(100)				

LOD = lack of data, SIR = Society of Interventional Radiology, USgHIFU = ultrasound-guided high-intensity focused ultrasound.

Discussion

This systematic review of seven cohort studies suggests that HIFU is an effective and safe treatment option for AWE. USgHIFU treatment led to reduced pain scores and lesion size, was free of major complications and had a pooled recurrence rate of 12.8 %. Compared to surgical excision pooled results showed no significant differences in pain scores, complications and recurrences after USgHIFU. However, it is important to note that there are only a few studies available and many of the included studies had limitations in their methodological quality, which could affect the overall strength of the conclusions and calls for cautious interpretation of the results.

Interpretation

NPV%

Of the three studies for which NPV could be calculated, one study reported a mean NPV%>100 %, i.e. the ablated volume was larger than the AWE lesion volume, mimicking wide surgical excision [23]. AWE lesions are infiltrating lesions. To ensure complete removal of the lesion, a margin-free zone of 5–10 mm is maintained during surgical excision [3,7]. This reasoning could apply to USg-/MRgHIFU as well since microscopic infiltrating endometriosis cells are often not visible on US of MRI and NPV > 100 % may ensure ablation of these microscopic infiltrating cells too [30–32]. It can therefore be argued that this should be pursued in all cases if safely possible by avoiding structures such as the rectus abdominis muscle, bladder or bowel. How large these margins

should be for USg-/MRgHIFU is unknown and should be investigated in future studies.

Reduction in pain scores and lesion size

All five studies reporting on pain scores showed a reduction in pain scores up to twelve months after USgHIFU treatment and three comparative studies showed no statistical difference in pain scores compared with surgical excision. According to Zhang et al. [33] and Benedetto et al. [3], 86.9 % of AWE patients initially seek medical attention for (cyclic) pain complaints, which in 61.4 % of patients are mild to intermediate (VAS1-6) and in 10.8 % severe (VAS ≥ 7). Therefore, reduction in pain scores can be seen as the most impactful outcome for patients and is thus one of the most important outcome measures to determine effectiveness of treatment. Furthermore, complete disappearance of the lesion is not required to achieve significant reduction in pain scores as shown by Luo et al. [26]. At six months, they showed a reduction in lesion volume of 52.5 ± 11.1 % with a corresponding reduction in pain scores of 5.0 ± 2.5 points. All studies that reported lesion size reported a sustained reduction or even disappearance of the lesion up to twelve months after treatment. Longer-term follow-up is needed to determine whether this reduction in size will eventually lead to disappearance, stabilization or regrowth of these lesions and how this correlates with NPV%, pain complaints and reintervention rates.

Side effects and complications

This study shows that side effects, although self-limiting, are relatively common after USgHIFU treatment. A possible explanation is that

AWE lesions are located just underneath the skin, resulting in more thermal injury in the near field, and are often located in a surgical scar consisting of avascular fibrous scar tissue that readily absorbs ultrasound energy, together causing local edema and pain complaints. [34] In addition, there is possible underreporting of side effects after surgical interventions, as none of the included comparative studies reported swelling or pain after surgery although these are standard, natural postsurgical side effects since incision causes an inflammatory response and activates nociceptive pain receptors, manifesting as pain, heat, erythema and swelling. [35] Future comparative studies should include detailed documentation of side effects and complications.

Recurrence and reintervention rates

In other HIFU indications, e.g. uterine fibroids and adenomyosis, higher NPV% rates correlate with better clinical outcomes [36,37]. As shown in by Hu et al. [29], this may also apply to the treatment of AWE. Significantly fewer recurrences were observed in the NPV%>50 % group compared to the NPV%<50 % group. However, the true predictive value of NPV% for clinical outcomes and at what threshold cannot be concluded from this review, since only a small number of studies reported NPV% and it had a wide range (58–177 %). Therefore, NPV% and its relationship with clinical outcomes should be included in future research.

USgHIFU vs. MRgHIFU

Only two case reports have been published on MRgHIFU for AWE, which were not included in this review. However the results of this review are likely to be applicable to MRgHIFU as well, since the HIFU technique is identical, only the visualization modality differs [9,38,39] MRgHIFU may even be favorable since MRI is the superior imaging modality for AWE due to its advantage of clear delineation and optimal (soft) tissue characterization together with the infiltrative nature of AWE lesions [30–32] Accurate imaging allows for precise treatment planning and real-time visualization of the AWE lesion increasing the likelihood of complete lesion ablation during treatment, subsequently improving clinical outcomes, and reducing recurrence rates.

Strengths and Limitations

This is the first systematic review on HIFU treatment of AWE in the 16 years that AWE has been treated with HIFU [28]. A thorough literature search was conducted and articles were critically reviewed and assessed for LoE and methodological quality. Side effects and complications were distinguished to make it applicable to clinical practice.

This review is limited by the lack of large, prospective or randomized studies. We could only include few retrospective cohort studies with small numbers of patients, low LoE and only one study of good methodological quality. Furthermore, recurrence rates were rarely reported and reintervention rates were not reported, although these rates are important information to support the implementation of this treatment in clinical practice. SDs often had to be estimated and NPV% were calculated based on mean NPV and mean pre-treatment lesion volume. Lesion size was often measured by different imaging modalities, resulting in less accurate values of NPV% and lesion size reduction.

Future perspectives

To demonstrate the (cost-)effectiveness of USg-/MRgHIFU compared to current standard of care, a well-structured, high-quality randomized controlled trial comparing USg-/MRgHIFU with surgical excision should be conducted, with appropriate reporting of NPV%, pain scores, side effects, complications, recurrences and reinterventions. In USg-/MRgHIFU treatment, the same imaging modalities should be used to determine AWE lesion size before USg-/MRgHIFU and on follow-up, preferably MRI, and complete ablation should be pursued. The need for ablated margins around the AWE lesion, NPV%>100 %, should be

investigated, with recurrence as outcome.

Ideally, the (cost-)effectiveness of MRgHIFU compared to USgHIFU should be included, as MRgHIFU seems to have a unique place in AWE treatment, but it remains unclear whether MRgHIFU is cost-effective compared to USgHIFU. This is important, as clinical adoption depends on (cost-)effectiveness in our current healthcare landscape of rising costs and consequent social burden.

Conclusions

The results of this systematic review suggest that USg-/MRgHIFU is an effective treatment option for AWE, leading to reduced pain scores, without major complications and does not appear to be inferior to surgical excision. However, studies on this topic are scarce with no comparative cohort studies of good methodological quality and no randomized controlled trials, so the results of this review should be interpreted with caution. A well-structured high-quality randomized controlled trial with well-defined target volumes and NPV% aiming at total ablation should be performed, to provide conclusive evidence on the (cost-)effectiveness of USg-/MRgHIFU for AWE compared to the current standard of care.

CRediT authorship contribution statement

Elisabeth R. Knorren: Conceptualization, Methodology, Investigation, Formal analysis, Visualization, Writing - original draft. **Larissa A. de Ridder:** Investigation, Formal analysis. **Ingrid M. Nijholt:** Writing - review & editing, Visualization, Methodology, Formal analysis, Conceptualization. **Jeroen R. Dijkstra:** Writing - review & editing, Visualization. **Manon N.G.J.A. Braat:** Writing - review & editing, Visualization. **Judith A.F. Huirne:** Writing - review & editing, Visualization. **Martijn F. Boomsma:** Writing - review & editing, Visualization, Conceptualization. **Joke M. Schutte:** Writing - review & editing, Visualization, Conceptualization.

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Declaration of competing interest

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Appendix A. Supplementary material

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ejogrb.2024.03.029>.

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